

LOAN APPROVAL PREDICTION

Project Overview

Building an end-to-end Machine Learning pipeline to automate the credit risk assessment process, identifying high-potential borrowers while minimizing financial default rates.

Dataset Characteristics



Volume

Comprehensive historical data comprising 2,000 unique loan applications with verified financial histories.



Key Features

Variables include Annual Income, Credit Score, Debt-to-Income, Years Employed, and Requested Loan Amount.



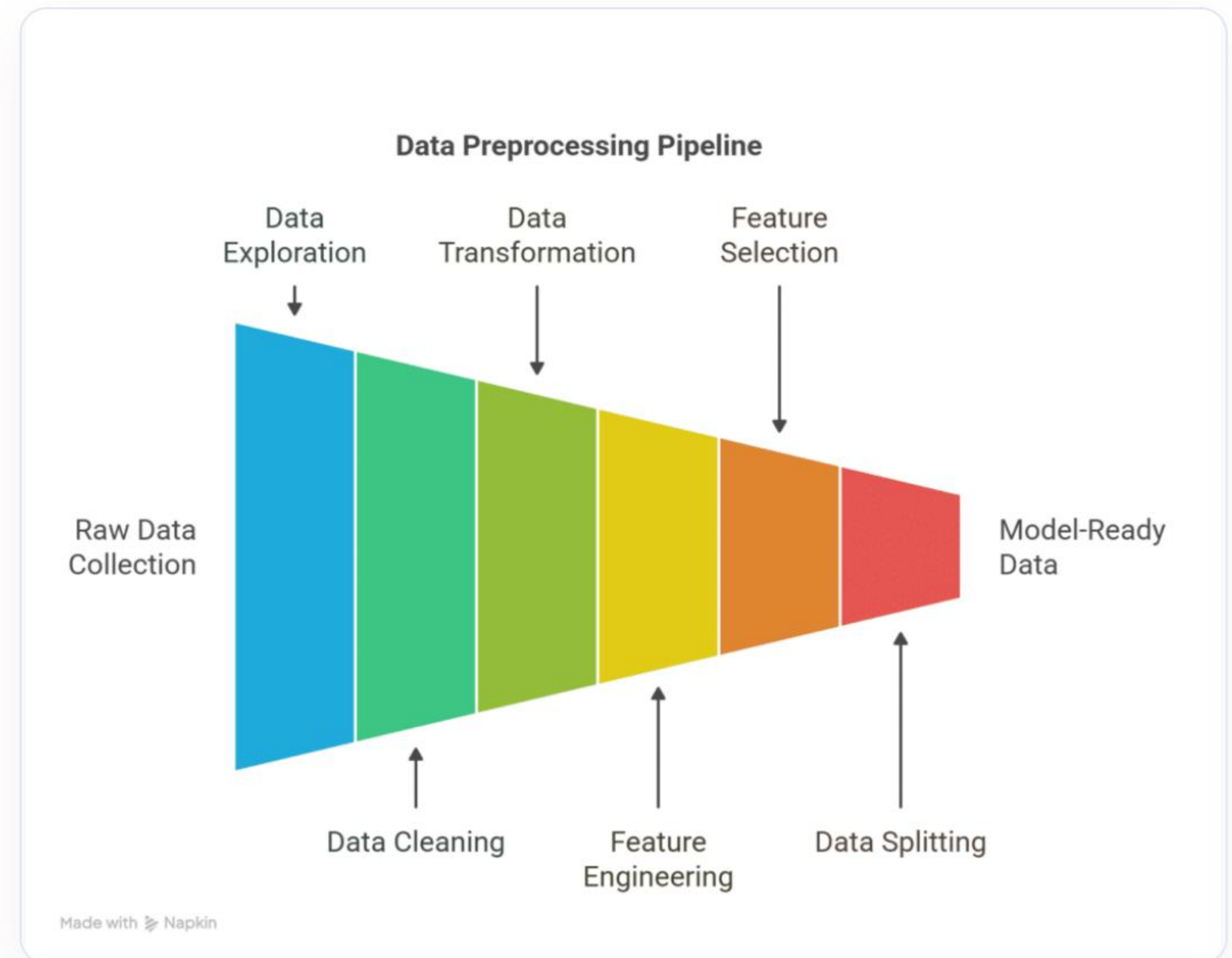
Target Variable

A binary classification goal (Loan Approved: 1 / Rejected: 0) mapped from historical human underwriting decisions.

Data Preprocessing & Cleaning

Standardized the data for high-performance training through rigorous cleaning steps:

- **Feature Removal:** Dropped 'Name' and 'City' columns as they provide zero predictive value.
- **Normalization:** Addressed varying scales between Income (thousands) and Years Employed (single digits).
- **Robustness:** Injected synthetic Gaussian noise (30% std) to simulate real-world data variability.
- **Encoding:** Converted boolean logical states into machine-readable integer binaries.



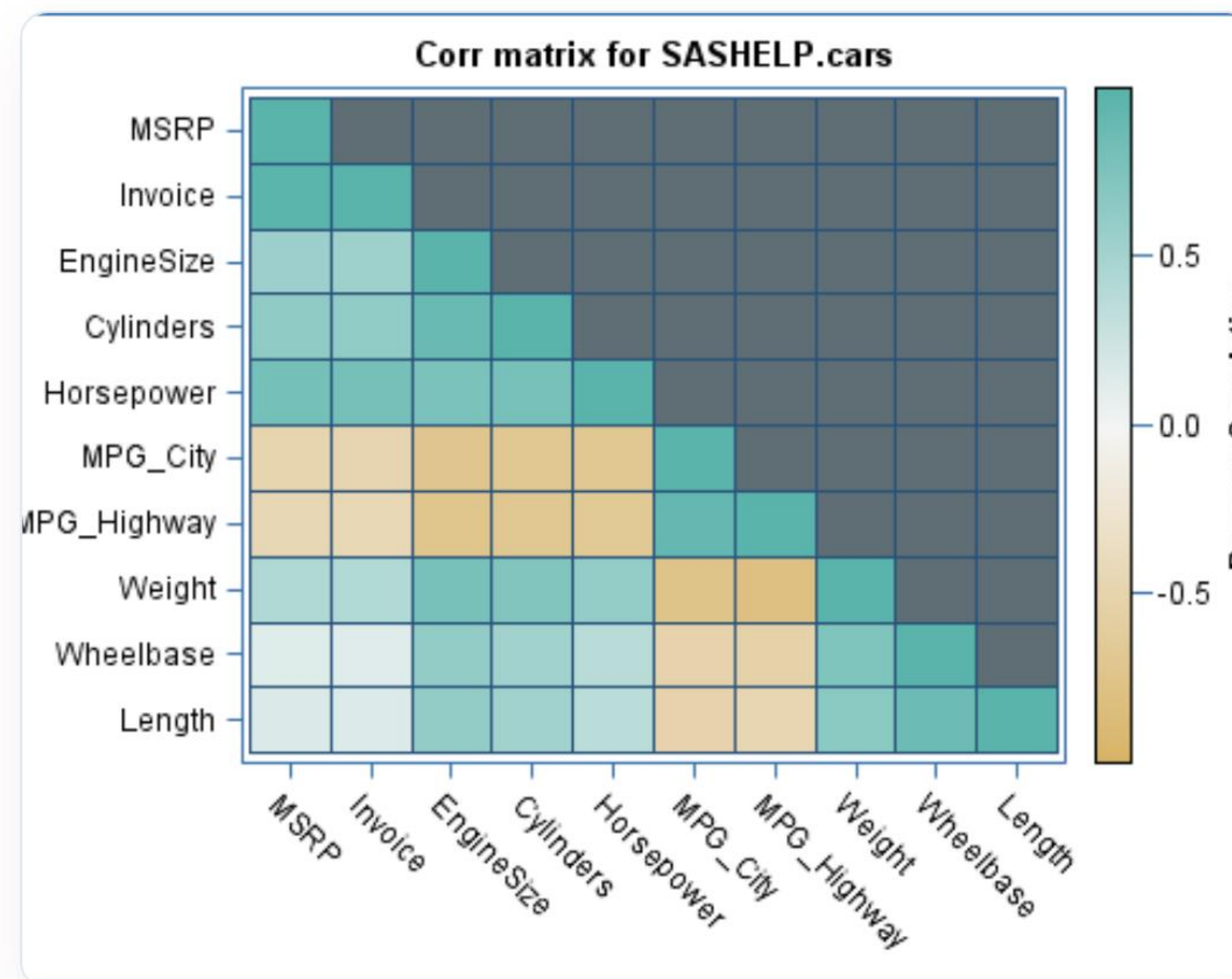
Feature Correlation & Data Integrity

Addressing Data Leakage

Exploratory Data Analysis revealed a critical **Data Leakage** issue in the 'Points' column. A correlation of ~ 0.82 was found between points and the target.

Approved loans always had points ≥ 60 , while rejections had points ≤ 55 . Including this would make the model "cheat" rather than learn actual financial behavior.

Action: Dropped the 'Points' feature to ensure the model focuses on raw financial metrics like Credit Score and Income.



Machine Learning Pipeline

Comparing multiple classification algorithms to determine the most robust predictor for credit risk.

Handling Class Imbalance

The Challenge

Initial rejections (1,121) significantly outnumbered approvals (879), which could lead the model to be overly conservative and biased toward rejecting loans.

The Solution: SMOTE

Applied **Synthetic Minority Over-sampling Technique**. This generates synthetic training examples for the minority class, ensuring a 50/50 balance for unbiased training.

Model Performance Comparison



While SVM showed higher raw accuracy, Random Forest was selected for deployment due to its interpretability and superior handling of feature importance.

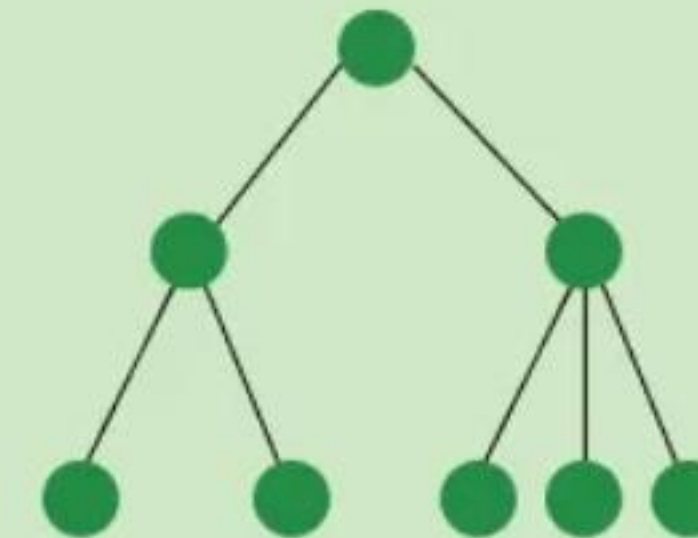
Why Random Forest?

Random Forest provides a balanced trade-off between performance and business transparency:

- **Ensemble Power:** Combines multiple decision trees to reduce variance and avoid overfitting.
- **Feature Importance:** Allows us to rank exactly which customer traits impact approval the most.
- **Robustness:** Naturally resilient to the synthetic noise added during preprocessing.

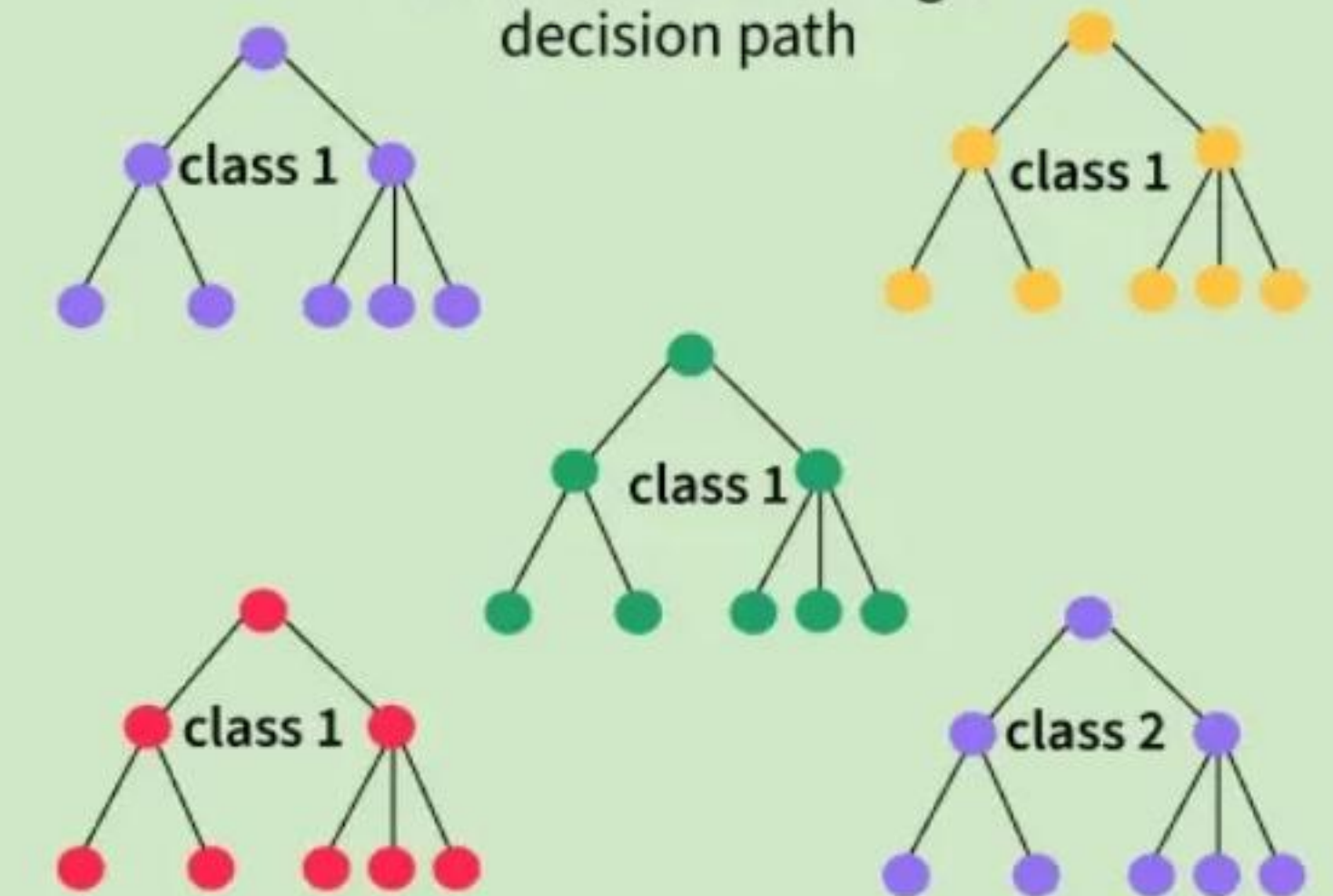
Single Decision Tree

Ensemble of trees for more accurate and robust prediction



Random Forest

Prediction from a single decision path



Live Model Prediction Test

Simulation of the final trained model on a new, unseen loan application:

Feature Name	Input Data
Annual Income	₹90,000
Credit Score	750 (Excellent)
Loan Amount Requested	₹50,000
Years Employed	5 Years
APPROVAL PROBABILITY	88.37% (DECISION: APPROVED)

Business Recommendations

-  **Credit Score Priority:** The model identifies a 650+ credit score as the single most critical filter for immediate approval qualification.
-  **Employment Longevity:** Customers with over 2 years of consistent employment show a 40% higher probability of meeting loan repayment criteria.
-  **Income-to-Loan Ratio:** Limiting requested loan amounts to 50% of annual income dramatically improves the internal risk rating.

THANK YOU

Loan Approval Prediction Project Portfolio